

Prathmesh Jain

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EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 – Present

Full Stack Engineer (Research & Innovation, CTO Team, BFSI Americas)

Mumbai, IN

- Built asynchronous workflow systems using **FastAPI, React, and LLM integrations** to automate compliance screening, document processing, and payment repair operations across multiple internal platforms.
- Developed concurrent backend pipelines for Google Custom Search API calls, web crawling, document extraction, and verification tasks, enabling long-running processes to execute independently while improving overall system responsiveness.
- Implemented agent-driven workflows using LLM-based orchestration for adverse media analysis, credit memo generation, and payment repair flows involving multi-step validation and tool-based reasoning.
- Designed real-time streaming interfaces using Server-Sent Events to stream workflow progress, intermediate outputs, and asynchronous task updates directly to frontend applications.
- Built high-throughput extraction and verification pipelines using Playwright-based crawling, multithreaded document processing, and LLM-assisted analysis across large unstructured datasets and financial reports.
- Developed persisted and resumable workflows with rollback handling, validation checkpoints, and human review stages to support long-running financial spread generation and transaction repair operations.
- Implemented reliability patterns including retries, fallbacks, circuit breakers, exponential backoff, and graceful degradation across AI-based systems to improve fault tolerance and user experience during partial failures.
- Worked across frontend and backend systems using React, FastAPI, PostgreSQL, Zustand/Redux, RBAC, authentication flows, file uploads, and Docker-based deployments as a core contributor across multiple projects.

PERSONAL PROJECTS

Transcribe – Lecture Transcription Workspace | *React.js, FastAPI, Redis, PostgreSQL*

[Live Demo](#)

- Developed an AI-powered lecture transcription platform that converts long-form spiritual discourses into structured, searchable, and downloadable PDF transcripts.
- Built responsive frontend interfaces for authentication, file uploads, transcription history, live processing status, and document downloads.
- Implemented an asynchronous processing pipeline using Redis queues, FFmpeg-based media conversion, silence detection, and chunked processing for scalable long-form transcription.
- Optimized transcription quality for Sanskrit shlokas (verses) and multilingual pravachans (discourses) using Gemini, followed by AI-powered transcript aggregation into structured HTML and automated PDF generation for 50+ active users.

ExpenseGauge – Expense Tracking App | *React Native, Node.js, MongoDB*

[Live Demo](#) | [GitHub](#)

- Designed and shipped a cross-platform, offline-first expense management application with real users and a downloadable Android APK.
- Implemented optimistic UI updates and background sync queues to ensure seamless usage during network outages.
- Implemented secure authentication using short-lived access tokens and rotating refresh tokens for seamless session continuity.
- Developed admin dashboards for user management, transaction audits, and balance adjustments.
- Built role-based backend services supporting expense analytics with monthly trends and category-wise visualizations.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Backend: Node.js, Express.js, FastAPI, Flask

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS

Database: MongoDB, PostgreSQL

Tools: Docker, Redis, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Mumbai, IN

Bachelors in Computer Engineering

May 2024

GPA: 9.2 / 10